

Comparing Seasonal Adjustment Methods: X-11 vs. SEATS

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Seats vs. X-11

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The empirical method [X-11] and the model-based method [Seats] are sometimes thought of as rivals. But they are only rivals in the same sense that the two sexes are rivals. In both cases, isolation is necessarily sterile, while interaction can be fruitful.
(Box, Hillmer, and Tiao, 1978)

History of Seats and X-11

- Origins: X-11 (U.S. Census 1965) and SEATS (Bank of Spain; Burman (1980)) 
- X-11 (Shiskin 1967) → X-11-ARIMA/88 (Dagum 1988) → X-12-ARIMA (1998) → X-13ARIMA-SEATS
- TRAMO/SEATS: Gómez and Maravall: TRAMO (Reg-ARIMA) and SEATS (ARIMA signal extraction)
- Adoption: X-11 was standard for decades; SEATS evaluated as a supplement in the 2000s: X-12-SEATS enabled side-by-side comparisons and common diagnostics

Features of Seats

	Seats
Advantage	• Attempts to only adjust seasonal series • Generally no overadjustment  • Drawing inference is more straight-forward • Model-based prediction of components • Infinite amount of filters • Can often handle breaks in seasonality
Disadvantage	• If, by mistake, a non-seasonal ARIMA model is found $\rightsquigarrow$ no adjustment • Depends on suitability of ARIMA models

Features of X-11

	X-11
Advantage	• User can influence results • Rather robust against outliers • Can handle seasonal heteroskedasticity
Disadvantage	• Limited number of filters • Non-model based choice of filters



Studies comparing Seats and X-11

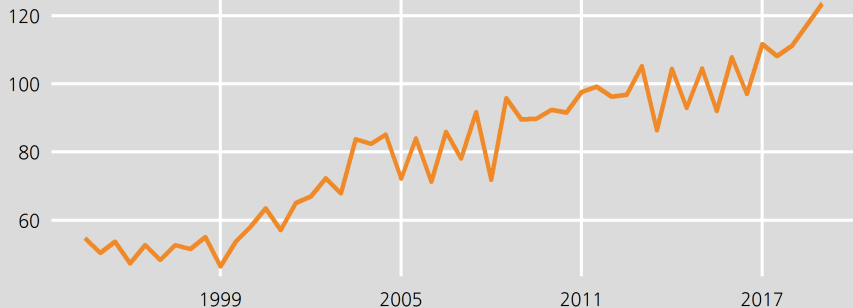
- Seats tends to use longer filters (Tiller, Chow, Scott, 2007)
- Seats may be better for highly irregular series (Hood, Ashley, Findley, 2000)
- X-11 tends to be better for short series (ibid.; Mazzi, Savio, 2005)

Impact of model order for Seats

Badly adjusted

55.3 Turnover on camping grounds

2015=100 - Automatic model: ARIMA(1 1 0)



— Unadjusted figures — Seasonally adjusted

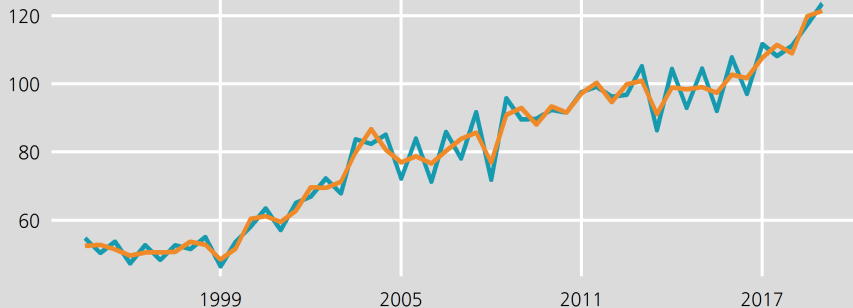
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Impact of model order for Seats

Well adjusted

55.3 Turnover on camping grounds

2015=100 - User-defined model: SARIMA(0 1 1)(0 1 1)



— Unadjusted figures — Seasonally adjusted

Deutsche Bundesbank

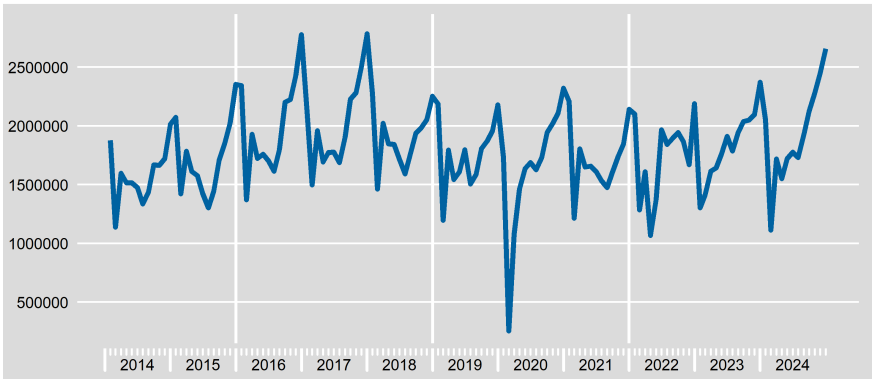
Model order

Take-away

- Think more about the model order used
- Check seasonality tests

Revisions (I/V)

China: Car sales of Passenger cars in domestic sales



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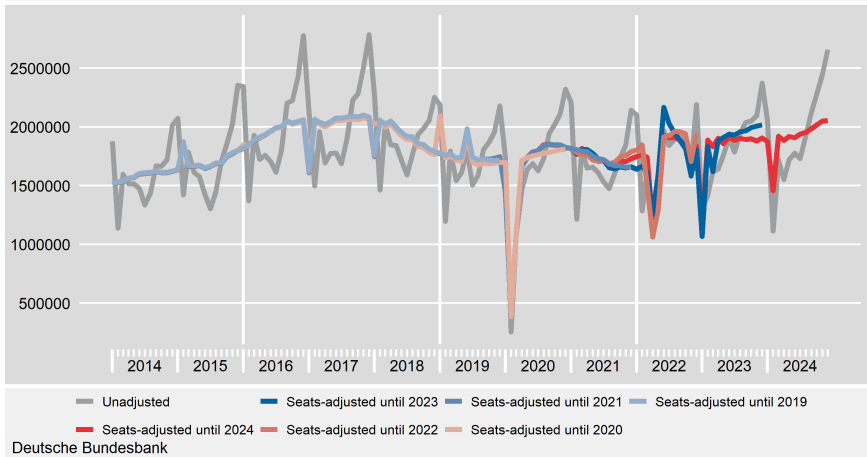


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Revisions (II/V)

China: Car sales of Passenger cars in domestic sales

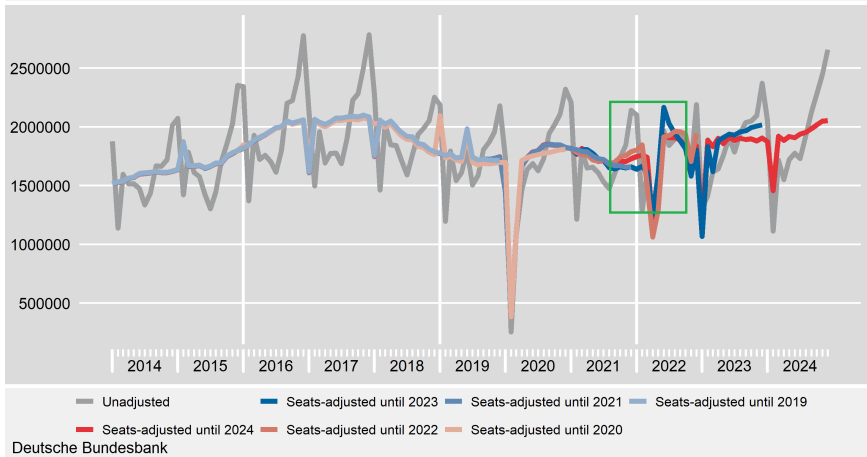
Revision analysis for Seats



Revisions (III/V)

China: Car sales of Passenger cars in domestic sales

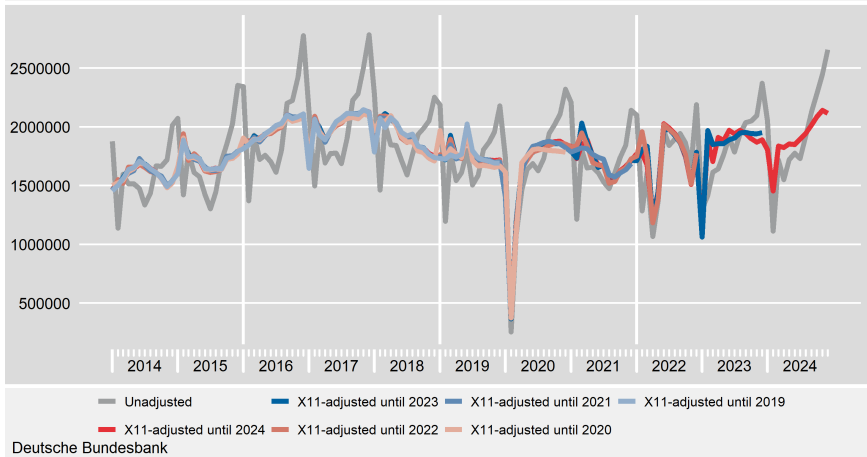
Revision analysis for Seats



Revisions (IV/V)

China: Car sales of Passenger cars in domestic sales

Revision analysis for X-11



Revisions (V/V)

Take-away

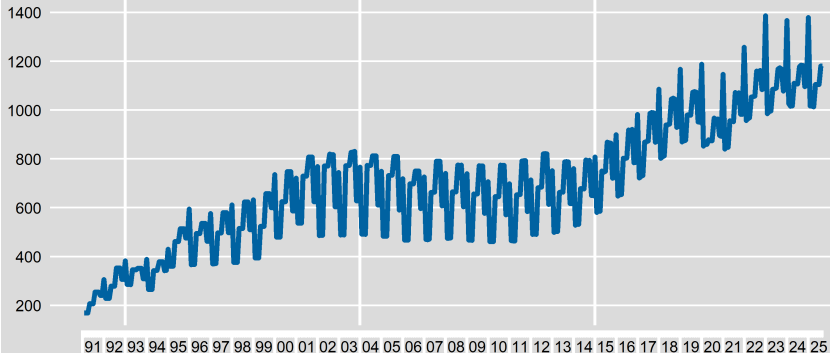
- Seats may react strongly to irregular occurrences
- Outlier adjustment is very important for Seats

Seasonal breaks (I/V)

Adjustment with X-11

Balance of payments - Primary income: compensation of employees - Debit

€ million



— Original series

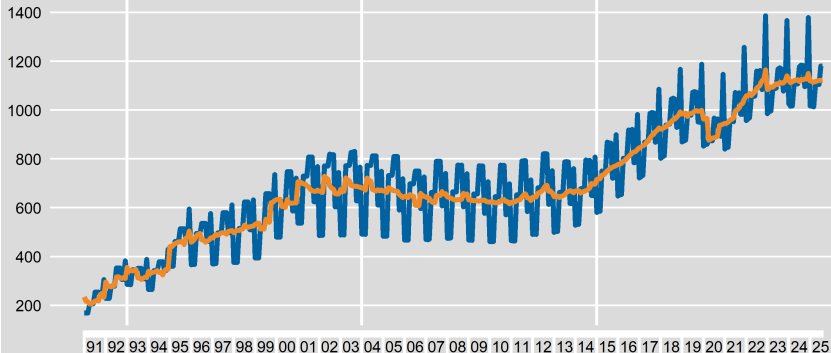
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Seasonal breaks (II/V)

Adjustment with X-11

Balance of payments - Primary income: compensation of employees - Debit

€ million



— Original series — Seasonally adjusted (X-11)

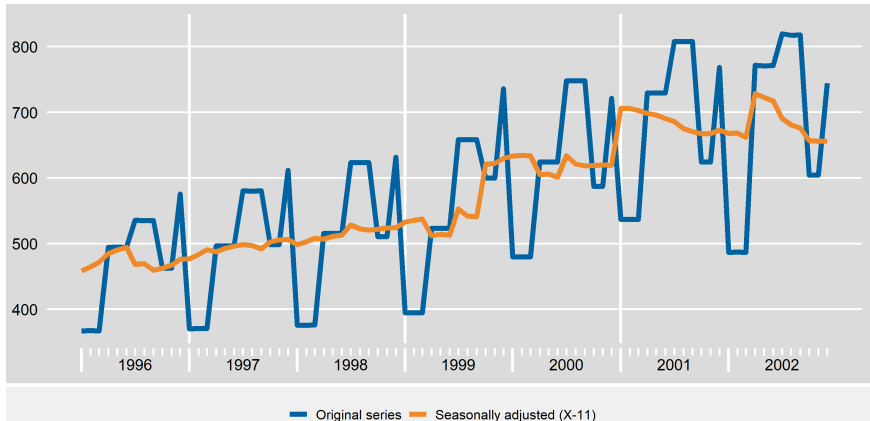
Deutsche Bundesbank

Seasonal breaks (III/V)

Adjustment with X-11

Balance of payments - Primary income: compensation of employees - Debit

€ million



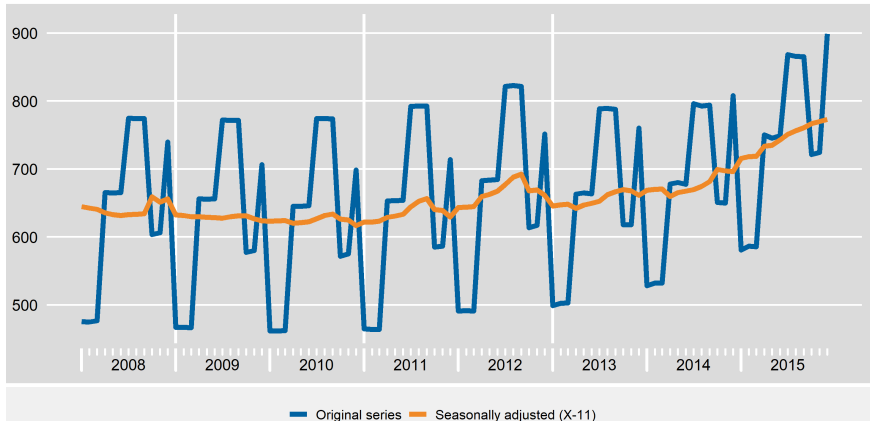
Deutsche Bundesbank

Seasonal breaks (IV/V)

X-11 vs. Seats

Balance of payments - Primary income: compensation of employees - Debit

€ million

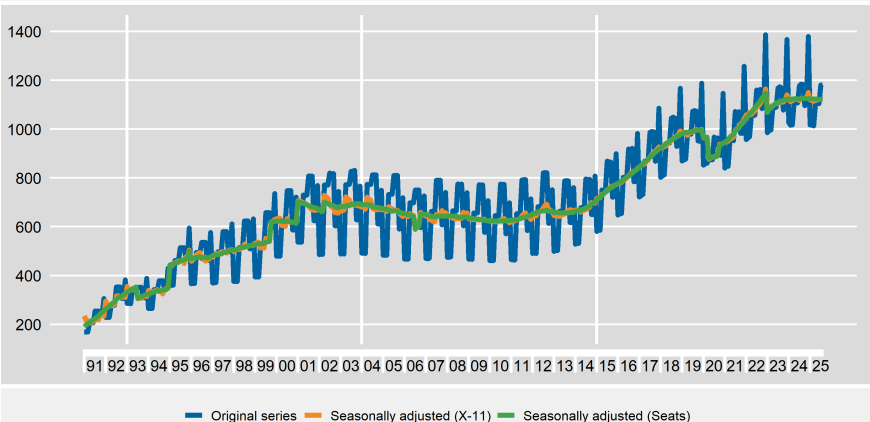


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Seasonal breaks (V/V)

Balance of payments - Primary income: compensation of employees - Debit

€ million

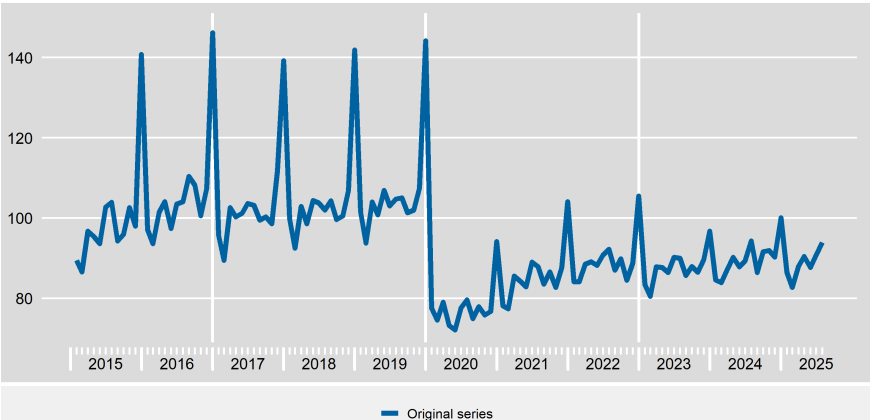


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Seasonal and method breaks (I/V)

Production in Service - Section L - Real estate activities

2015=100



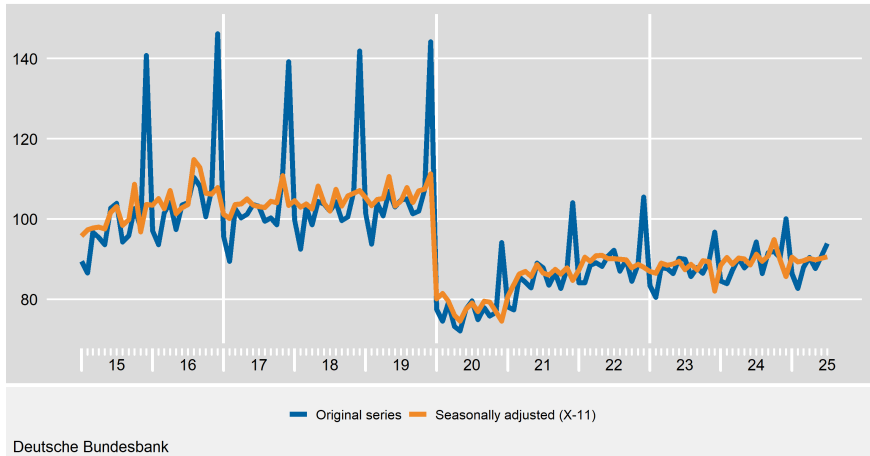
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Seasonal and method breaks (II/V)

Adjustment with X-11

Production in Service - Section L - Real estate activities

2015=100

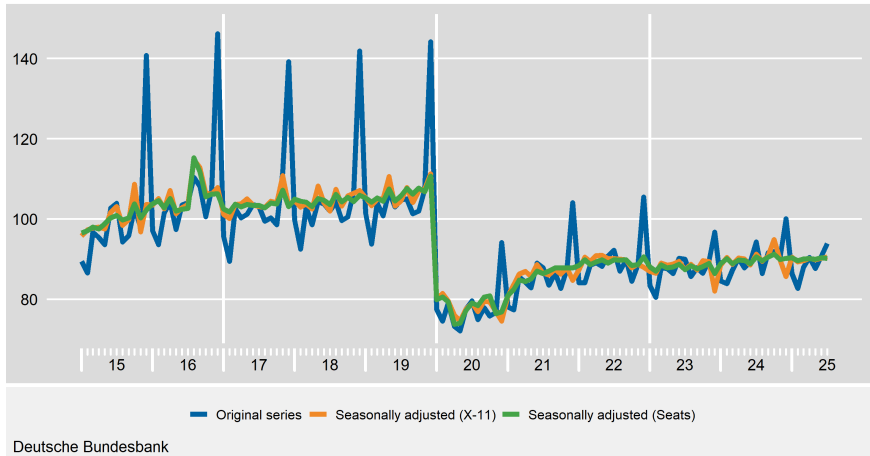


Seasonal and method breaks (III/V)

X-11 vs. Seats

Production in Service - Section L - Real estate activities

2015=100

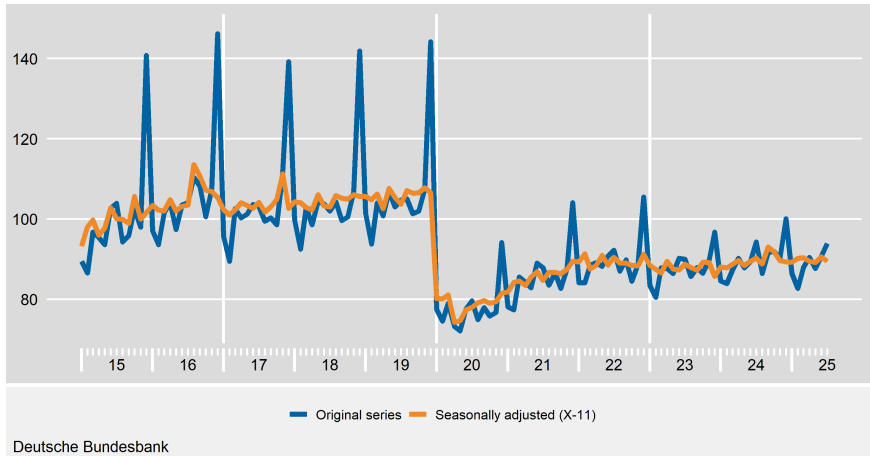


Seasonal and method breaks (IV/V)

Adjustment with X-11 and Seasonal outliers

Production in Service - Section L - Real estate activities

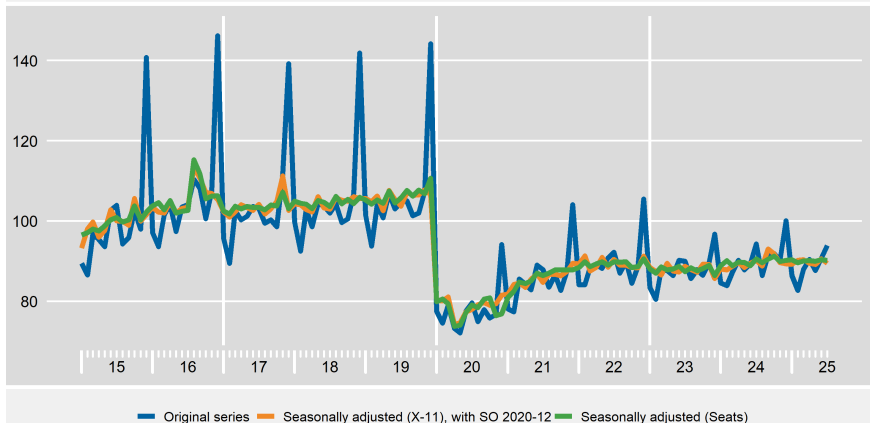
2015=100, with seasonal outlier in 2020-12



Seasonal and method breaks (V/V)

Production in Service - Section L - Real estate activities

2015=100



Deutsche Bundesbank

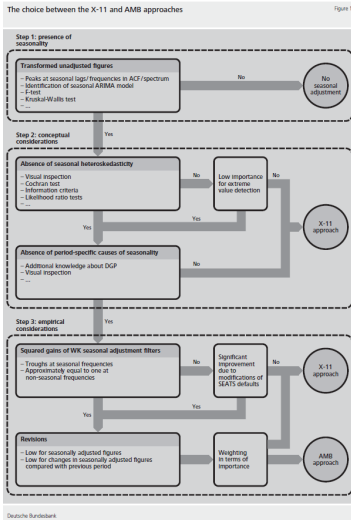
Breaks in series

Take-away

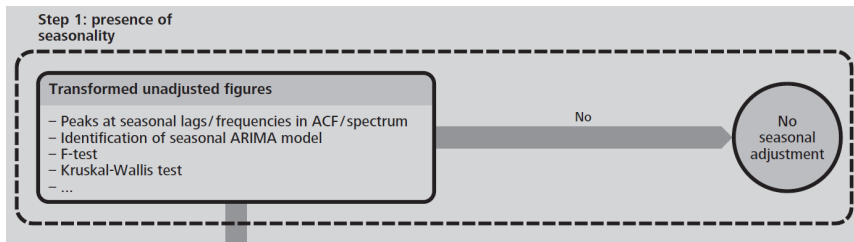
- With enough years of observations, Seats tends to be better at modelling seasonal breaks
- Seasonal outliers may improve the adjustment of X-11
- Alternatively: splitting the series may improve the adjustment of X-11

Algorithm: Seats or X-11 (I/IV)

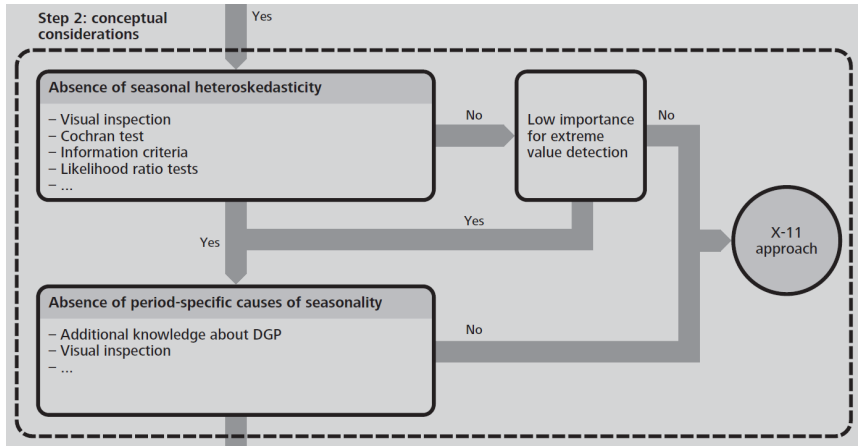
Webel (2016)



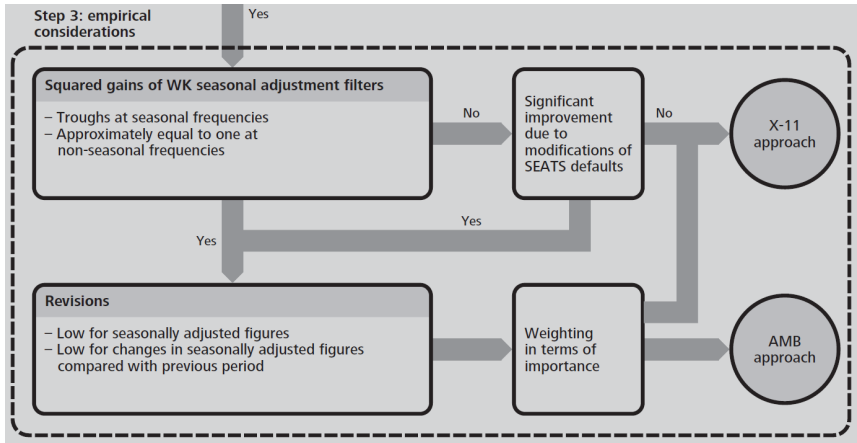
Algorithm: Seats or X-11 (II/IV)



Algorithm: Seats or X-11 (III/IV)



Algorithm: Seats or X-11 (IV/IV)



Summary: Seats vs. X-11



Seats vs. X-11

The empirical method [X-11] and the model-based method [Seats] are sometimes thought of as rivals. But they are only rivals in the same sense that the two sexes are rivals. In both cases, isolation is necessarily sterile, while interaction can be fruitful.

(Box, Hillmer, and Tiao, 1978)

Seats

-  Box, G.E.P., Hillmer, S.C. & Tiao, G.C. (1978). Analysis and Modeling of Seasonal Time Series. In Seasonal Analysis of - Economic Time Series, ed. A. Zellner. Washington. D.C.: U.S. Dept. of Commerce. Bureau of the Census. 309-334.
-  Dagum, Bianconcini (2016). Seasonal Adjustment Methods and Real Time Trend-Cycle Estimation. Springer: Bologna.
-  Gómez, Maravall (2001). Seasonal Adjustment and Signal Extraction in Economic Time Series. In: Peña, Tiao, Tsay (eds). A Course in Time Series Analysis. Wiley: New York. 202-247.
-  Maravall (1995). Unobserved Components in Economic Time Series. In: Pesaran & Wickens (eds), The Handbook of Applied Econometrics Vol 1. Blackwell Publishing: Oxford. 12-72.

Seats vs. X-11

-  Hood, C.C., Ashley, J.D. & Findley, D.F. (2000). An empirical evaluation of the performance of TRAMO/SEATS on simulated series.
-  Mazzi G. & Savio G. (2005). The Seasonal Adjustment of Short Time Series. European Commission Working Papers and Studies.
-  Tiller, R. T., Chow, D., & Scott, S. (2007). Empirical Evaluation of X-11 and Model-Based Seasonal Adjustment Methods.
-  Webel, K. (2016). A data-driven selection of an appropriate seasonal adjustment approach. Deutsche Bundesbank Discussion Paper Series 07/2016.